

Multi-Sensor Monitoring for Cultural Heritage Preservation: Overcoming Single-Modality Limits with RGB-IR Sensor Fusion

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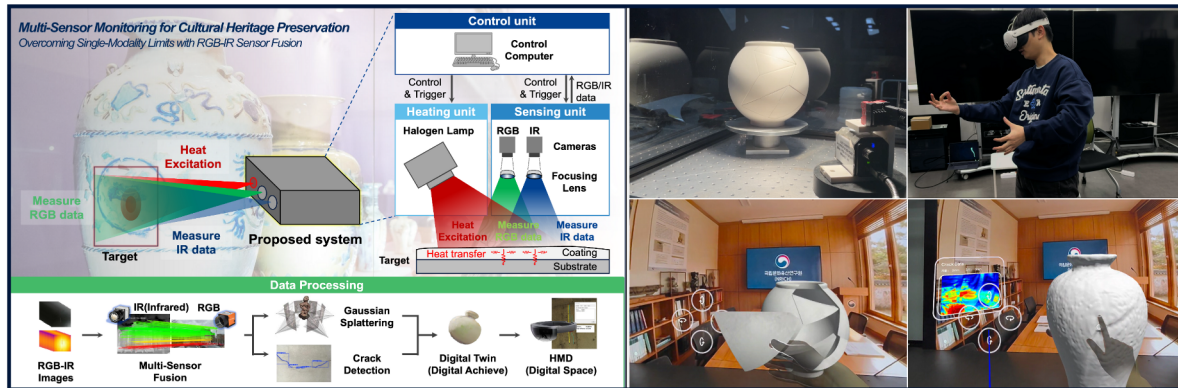


Figure 1: Overview of Multi-Sensor Monitoring System.

ABSTRACT

Cultural heritage inspection often relies on RGB images and single-sensor 3D capture, which limits subsurface defect evidence and long-term condition tracking. A multi-sensor digital-twin pipeline is presented for non-contact diagnosis and time-series archiving. The pipeline synchronizes RGB and infrared thermography, fuses multi-view multispectral observations with RGB-IR Gaussian Splatting to build an as-is 3D restoration model, and stores repeated scans for longitudinal comparison. An AR interface overlays the multispectral digital twin on the physical artifact for interactive defect exploration and reporting. Experiments across three crack phases validate crack detectability and reconstruction quality (RGB PSNR 32.12 dB; IR 36.03 dB).

Index Terms: Cultural heritage preservation; Digital twin; Non-destructive testing; Multispectral sensing; Gaussian Splatting; Augmented reality.

1 INTRODUCTION

Cultural heritage assets deteriorate due to material aging, environmental exposure, and unexpected events, yet restoration decisions often must be made on site under strict time and resource constraints. Standard documentation generates heterogeneous outputs—photographs, sensor-based visualizations, and 3D models—that are typically interpreted separately, increasing cognitive load and slowing on-site decision-making; field variability such as changing illumination, restricted accessibility, and occlusions further degrades repeatability and cross-visit comparability, reducing confidence in defect localization and long-term tracking. Augmented reality can mitigate these limitations by aligning and visualizing 3D restoration models directly on the physical artifact,

integrating surface appearance, thermal defect cues, and geometry within a single spatial reference for intuitive exploration of surface and (sub)surface damage, severity, and temporal change, as well as faster evidence sharing and clearer reporting. A shared XR workspace extends this capability to collaborative judgment by enabling on-site and remote experts to co-annotate and discuss damage within a common spatial frame and coordinate restoration priorities through shared spatial awareness and joint attention [2, 3]. Building on this premise, we present a multi-sensor digital-twin system with three contributions: (1) a non-contact, non-destructive multi-sensor inspection setup; (2) RGB-IR Gaussian Splatting (GS) for 3D reconstruction and (sub)surface defect monitoring using co-registered multispectral channels; and (3) a time-series digital archiving framework that enables longitudinal comparison, condition monitoring, and timely maintenance planning. A Unity-based AR interface overlays the multispectral digital twin onto the physical artifact, supporting real-time interactive defect exploration and clear reporting for both experts and non-experts.

2 MULTI-SENSOR DIGITAL ARCHIVING SYSTEM

The proposed system combines non-contact inspection hardware with a software pipeline for multispectral registration, defect visualization, 3D digital-twin construction, and time-series condition monitoring. It supports repeated inspections under variable site conditions and provides a consistent spatial reference for AR visualization and reporting (Figure 1).

2.1 Hardware Configuration

The hardware setup synchronously captures RGB and long-wave infrared (LWIR) data under controlled thermal excitation. A rigid RGB-IR camera rig maintains fixed relative geometry for stable cross-modal registration, while a halogen lamp applies short, localized heating to elicit thermal responses that expose (sub)surface discontinuities. A control computer synchronizes triggering, excitation timing, and logging to ensure temporal alignment and repeatability. Motorized rotation and vertical positioning enable multi-view acquisition without direct handling, and each measurement

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is stored with viewpoint and timing metadata for consistent reconstruction, defect mapping, and time-series archiving.

2.2 Software Configuration

Software configuration coordinates synchronized acquisition, multispectral calibration/registration, defect-cue extraction, multispectral 3D reconstruction, and AR deployment. RGB and IR sequences are co-registered to establish geometric correspondence across modalities and viewpoints, allowing IR-derived defect cues to be localized on surface appearance and geometry. Thermal-response analysis extracts crack-related signatures while suppressing non-defect variation from texture and environmental effects, and stores the resulting defect maps in the same spatial reference as the RGB observations. For digital-twin construction, camera poses are estimated with COLMAP [6], followed by per-modality GS reconstruction, with SuGaR regularization enabling mesh extraction and efficient interactive rendering [4, 5]. The spatially aligned RGB and IR reconstructions are fused into a multispectral digital twin, and time-indexed archives support longitudinal comparison across repeated inspections. Unity imports the twin and defect layers for AR overlay and interactive review, facilitating inspection communication and field reporting.

3 COLLABORATIVE XR RESTORATION GUIDANCE

Cultural heritage diagnosis often requires interpreting photographs, thermal images, and 3D models, and the workload grows when these modalities remain separated. A multispectral digital twin unifies these heterogeneous observations within a single spatial reference, reducing cognitive effort during defect localization and enabling consistent interpretation of surface geometry alongside (sub)surface defect cues. An AR workflow then overlays the multispectral twin directly onto the physical object, allowing on-site inspectors to connect evidence to real-world locations without manual cross-referencing across documents or screens [1]. Building on this alignment, a collaborative XR workflow with two Meta Quest headsets extends AR visualization into shared decision-making: a remote expert reviews the digital twin and time-series records in an immersive workspace to identify defect locations and severity, while an on-site operator views the same twin aligned through AR and carries out inspection or restoration actions under the expert's guidance. Shared coordinates and annotations anchor both users to identical defect locations, enabling spatially grounded instructions, rapid on-site confirmation, and reduced ambiguity during interventions. Through shared spatial awareness and joint attention, the workflow supports faster and more consistent judgments on restoration priority and maintenance actions [2, 3].

4 EXPERIMENTAL SETUP AND RESULTS

Evaluation utilized a synchronized RGB-IR acquisition setup controlled by a Raspberry Pi 5. The RGB stream was captured using an Allied Vision ALVUM 1800 U-240 at 1936×1216 pixels (15 Hz), while the infrared stream was acquired via a FLIR A65SC (LWIR) with 40 mK NETD at 640×480 pixels (30 Hz). Both sensors were positioned approximately 50 cm from the specimen, with a 1.5 kW halogen lamp providing thermal excitation to induce temperature contrast. The experiment tracked three crack progression phases—Phase A (intact), Phase B (initial formation), and Phase C (repair required)—under consistent conditions. Figure 2 summarizes the results, showing that the workflow detected all cracks with phase-dependent cues. The GS reconstruction achieved average PSNRs of 32.12 dB for RGB and 36.03 dB for IR, while the Unity-based AR visualization provided a unified spatial reference for intuitive defect exploration and reporting.

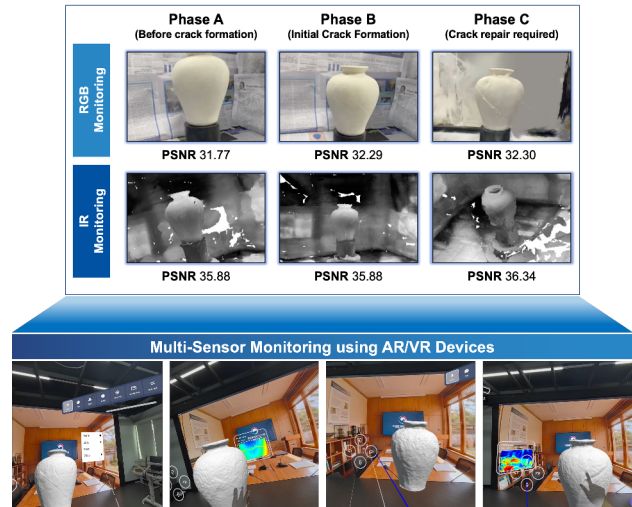


Figure 2: Phase-wise RGB-IR Gaussian Splatting results (PSNR) and Unity-based AR visualization for multi-sensor monitoring.

5 CONCLUSION

This work presents a multi-sensor digital-twin pipeline integrating non-contact RGB-IR inspection, multispectral Gaussian Splatting, and time-series archiving for (sub)surface defect monitoring. By unifying sensing, reconstruction, and AR within a single spatial reference, the pipeline enables practical and repeatable conservation workflows. An AR/XR interface further supports intuitive defect localization and remote expert guidance. Results demonstrate reliable crack detection and high-quality reconstruction, with future work targeting diverse artifacts and field efficiency validation.

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